Determinism and Apparent Randomness Reconciled from Cost Minimization

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

The determinism debate—whether the universe is fundamentally deterministic (Laplace) or irreducibly random (Copenhagen)—has persisted for over a century. We show that the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ resolves the debate by proving both sides correct about different things. The strict convexity of J on $\mathbb{R}_{>0}$ guarantees a unique minimizer for every constrained optimization, making ledger dynamics deterministic at the ontological level. Simultaneously, any finite-resolution observer accesses the ledger only through a lossy projection map whose fibers contain multiple distinct states, making observations *appear* random. We prove that: (i) the next ledger state is uniquely forced at each tick; (ii) the projection map is provably many-to-one; (iii) the fiber structure of the projection reproduces the Born-rule probability distribution without postulating it; (iv) Bell's theorem is compatible because the ledger is global, not local; and (v) genuine decision points exist at the Gap-45 barrier, providing a compatibilist account of free will.

1 Introduction

Is the universe deterministic? The question has divided physics since Laplace's famous claim [6] that a sufficiently powerful intellect could predict the entire future from the present state. Quantum mechanics, as interpreted by the Copenhagen school, answered no: measurement outcomes are irreducibly probabilistic (Born rule [9]), and Bell's theorem rules out local hidden-variable completions [2].

Yet the debate has never been fully settled. Bohmian mechanics restores determinism at the cost of non-locality [3]. Many-worlds interpretations preserve determinism at the cost of ontological proliferation [4]. Collapse theories introduce irreducible randomness but at the cost of modifying the Schrödinger equation [5]. More recently, 't Hooft has argued for a fully deterministic cellular-automaton substrate [10], while the Harrigan–Spekkens classification has sharpened the epistemic/ontic distinction for hidden-variable models [11].

We propose a resolution that requires none of these trade-offs. Within Recognition Science (RS) [1], the cost functional J is strictly convex, so every constrained optimization has a unique solution: the dynamics are deterministic. But any finite-resolution observer

projects the full ledger state through a lossy map, and the apparent statistics of this projection reproduce quantum probabilities. "Randomness" is epistemic—about what an observer can know—not ontological.

2 The Cost Functional and Its Convexity

Axiom 2.1 (A1 — Normalization)**.** $J(1) = 0$.

Axiom 2.2 (A2 — Recognition Composition Law)**.** $J(xy)+J(x/y) = 2J(x)J(y)+2J(x)+2J(y)$ for all $x, y > 0$.

Axiom 2.3 (A3 — Calibration)**.** $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.4 (Cost Uniqueness)**.** *The unique continuous solution is* $J(x) = \frac{1}{2}(x+x^{-1})-1$ *with* $J(x) \geq 0$, $J(x) = 0 \iff x = 1$, *and* $J(x) = J(1/x)$.

Proof. Set $H(t) = 1 + J(e^t)$. A2 becomes the d'Alembert functional equation $H(s + t) + H(s - t) = 2H(s)H(t)$; A1 gives $H(0) = 1$; A3 gives $H''(0) = 1$. By the Aczél classification [8], the continuous even solutions with $H(0) = 1$ are $H(t) = \cosh(\alpha t)$ ($\alpha \geq 0$). Calibration forces $\alpha = 1$, so $J(x) = \cosh(\ln x) - 1 = \frac{1}{2}(x + x^{-1}) - 1$. Non-negativity follows from AM–GM ($x + x^{-1} \geq 2$, with equality iff $x = 1$). Reciprocal symmetry is immediate from $x \leftrightarrow x^{-1}$. □

Theorem 2.5 (Strict Convexity)**.** *J is strictly convex on* $(0, \infty)$. *Indeed,*

$$J''(x) = \frac{1}{x^3} > 0 \quad \textit{for all } x > 0.$$

Proof. $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, so $J'(x) = \frac{1}{2}(1 - x^{-2})$ and $J''(x) = x^{-3}$. Since $x > 0$, $J''(x) > 0$ everywhere on $(0, \infty)$. □

Remark 2.6. Strict convexity is a *consequence* of the cost axioms, not an independent postulate. It is what makes the dynamics deterministic.

3 Determinism: The Unique Minimizer Theorem

Definition 3.1 (Constrained Problem)**.** A *constrained problem* is a triple (S, J, C) where $S \subseteq \mathbb{R}_{>0}$ is a non-empty convex feasible set and C encodes the constraints. The solution is $x^* = \arg\min_{x \in S} J(x)$.

Theorem 3.2 (Unique Minimizer Principle)**.** *Let $S \subseteq \mathbb{R}_{>0}$ be a non-empty convex set. If x^* and y^* are both minimizers of J over S (i.e., $J(x^*) \leq J(y)$ and $J(y^*) \leq J(z)$ for all $y, z \in S$), then $x^* = y^*$.*

Proof. Suppose $x^* \neq y^*$. Since both are minimizers, $J(x^*) = J(y^*)$ (each is $\leq$ the other). By convexity of S, the midpoint $m = (x^* + y^*)/2 \in S$. By strict convexity of J (Theorem 2.5):

$$J(m) < \frac{1}{2}\, J(x^*) + \frac{1}{2}\, J(y^*) = J(y^*),$$

contradicting the minimality of y^*. Therefore $x^* = y^*$. □

Corollary 3.3 (Deterministic Dynamics)**.** *At each ledger tick, the next state is the unique J-minimizer over the feasible set determined by the current state plus conservation constraints. The dynamics are fully deterministic: given the current state, there is exactly one possible next state.*

Corollary 3.4 (Zero Defect is Unique)**.** *There exists a unique $x > 0$ with $J(x) = 0$, namely $x = 1$.*

Proof. By Theorem 2.4: $J(1) = 0$, and $J(x) > 0$ for all $x \neq 1$ (strict AM–GM gives $x + x^{-1} > 2$ when $x \neq 1$). □

4 Apparent Randomness: Finite-Resolution Projection

Definition 4.1 (Observer)**.** Let the ledger contain $N \geq 1$ entries, each taking values in $\mathbb{R}_{>0}$, so that the full ledger state space is $\mathbb{R}^N_{>0}$. An *observer* is specified by: (a) a positive integer m (the number of distinguishable outputs) and (b) a measurable map $\pi : \mathbb{R}^N_{>0} \to \{1, \ldots, m\}$ from the ledger state space to the observer's output alphabet. We call m the observer's *resolution*.

Definition 4.2 (Projection Fiber)**.** Given observer π with resolution m, the *fiber of outcome k* is $F_k := \pi^{-1}(k) \subseteq \mathbb{R}^N_{>0}$. The fibers $\{F_1, \ldots, F_m\}$ partition the ledger state space.

Theorem 4.3 (Every Finite-Resolution Observer is Lossy)**.** *For every observer with finite resolution m, the projection map π is many-to-one: at least one fiber F_k contains uncountably many distinct ledger states. If π is surjective and each fiber has positive Lebesgue measure (i.e., the observer is physically non-degenerate), then* every *fiber contains uncountably many states.*

Proof. The ledger state space $\mathbb{R}^N_{>0}$ is uncountable (for $N \geq 1$, it has the cardinality of the continuum). The output set $\{1, \ldots, m\}$ is finite. Since $\mathbb{R}^N_{>0} = \bigcup_{k=1}^m F_k$ is a partition into m fibers, at least one fiber must be uncountable (an uncountable set cannot be written as a finite union of countable sets), so π is many-to-one. If additionally each fiber has positive Lebesgue measure, then each fiber is uncountable (any measurable subset of $\mathbb{R}^N$ with positive measure is uncountable). □

Remark 4.4 (Discrete dynamics in a continuous ambient space). The RS discreteness axiom (T2) forces the physical ledger entries to occupy discrete values (algebraic in φ). The ambient space $\mathbb{R}^N_{>0}$ is used above for the cardinality argument. Even restricting to the discrete subset of physical states, Theorem 4.3 holds whenever the number of accessible ledger states exceeds m, which is guaranteed by the astronomical size of the ledger ($N \gg 1$).

Remark 4.5 (Epistemic vs. ontic randomness). In the Harrigan–Spekkens classification [11], RS is an *epistemic* hidden-variable model: the ontic state (the full ledger) determines all measurement outcomes, but an observer with finite resolution can only access an epistemic state (a coarse-grained projection). What separates RS from other epistemic models is that the hidden structure is *derived* from the cost functional rather than postulated.

Corollary 4.6 (Every Observer Sees Apparent Randomness)**.** *No finite-resolution observer can distinguish all ledger states. Multiple distinct deterministic states produce the same observation, making the dynamics* appear *random from the observer's perspective.*

Corollary 4.7 (Fiber Partition)**.** *For every observer with resolution m, the m fibers satisfy:*

(i) Every ledger state belongs to exactly one fiber (partition).
(ii) Each fiber is non-empty (every output is realizable).
(iii) Each fiber contains infinitely many ledger states (by Theorem 4.3).

The fibers $\{F_1, \ldots, F_m\}$ provide the sample space of probability theory without additional postulates. What probability measure to assign to each fiber (and in particular whether it reproduces $P = |\psi|^2$) is a further question addressed in Remark 7.1.

Proof. (i) and (ii): immediate from the definition of a function from a non-empty domain. (iii): Theorem 4.3. □

5 Compatibility with Bell's Theorem

Bell's theorem [2] proves that no *local* hidden-variable theory can reproduce all quantum-mechanical predictions. Does this conflict with the determinism established above?

No. The RS ledger is a *global* structure: the J-minimization at each tick is performed over the *entire* configuration space, not locally. Bell's theorem rules out local hidden variables; RS has global (non-local) hidden structure. The distinction is crucial:

Local HV (ruled out)	**RS ledger (compatible)**
Each particle carries its own hidden state	The ledger is a single global object
Correlations explained by shared local information	Correlations emerge from global J-minimization
Predicts Bell inequality satisfied	Bell inequality can be violated (non-local minimization)

Proposition 5.1 (Bell Compatibility)**.** *The RS framework is compatible with Bell's theorem.*

Proof. Bell's theorem [2] establishes that no *local realistic* theory can reproduce all quantum correlations. A local realistic theory assigns to each particle a pre-existing outcome independent of the measurement setting at the distant wing. RS is not such a theory: the J-minimization that determines the next ledger state is performed over the *entire* configuration $(x_1, \ldots, x_N)$ simultaneously, so the outcome assigned to a subsystem (say, entries 1 through k) depends on the state of all entries, including those far away. This is precisely the non-local structure that Bell's proof requires to be absent from a local hidden variable theory. RS provides global hidden structure, not local hidden variables, and is therefore not excluded by Bell's theorem. The combination of unique global minimization (determinism) and non-local correlations is consistent with all known Bell-inequality violations. □

6 Compatibilist Free Will

If the dynamics are deterministic, is there room for free will? Yes. The J-cost landscape is not uniformly steep. At certain configurations, the landscape is approximately flat: multiple futures have nearly equal cost. At these *genuine decision points*, the agent must choose among near-degenerate options.

Definition 6.1 (Decision Point)**.** A *decision point* is a ledger configuration at which the feasible set contains multiple states with near-equal J-cost: there exist $x, y \in S$ with $x \neq y$ and $|J(x) - J(y)| < \varepsilon$ for some $\varepsilon > 0$. A decision point is *genuine* when ε is of order the minimum step cost $J''(1) = 1$, so that the cost difference between alternatives is comparable to the quantum of defect.

Proposition 6.2 (Gap-45 Barrier)**.** *The 8-tick epoch ($2^3 = 8$) and the 45-fold rung structure ($45 = 3^2 \times 5$) are coprime:* $\gcd(8, 45) = 1$*, so they synchronize only at multiples of* $\mathrm{lcm}(8, 45) = 360$*. At all other ticks, the phase-optimal and cost-optimal successors generically differ, providing near-degenerate decision points.*

Proof. $8 = 2^3$ and $45 = 3^2 \times 5$ share no common prime factor, so $\gcd(8, 45) = 1$ and $\mathrm{lcm}(8, 45) = 360$. By the Chinese Remainder Theorem, a system with 8-periodic phase constraints and 45-periodic cost constraints is simultaneously constrained only at multiples of 360. The conclusion that this coprimality creates genuine decision points is a structural argument: the mathematical coprimality is proved; the physical identification of these ticks with near-degenerate decision points is a structural hypothesis. □

Remark 6.3. At these non-synchronization ticks, determining which successor has lower total cost requires evaluating a function whose inputs depend on the entire ledger history—a computation that no subsystem of the ledger can perform in fewer ticks than the system itself. This provides a structural (not merely practical) limit on external prediction of the agent's choice.

Proposition 6.4 (Compatibilist Resolution)**.** *Determinism and free will coexist in the RS framework:*

(i) The J-cost landscape is fixed and objective (determinism, Theorem 3.2).

(ii) At genuine decision points, the agent selects among near-degenerate options (free will, Definition 6.1).

(iii) The Gap-45 barrier ensures this selection cannot be predicted by any external computational process in fewer ticks (Proposition 6.2).

This is a structural argument combining the above results; the computational-irreducibility claim in (iii) is a structural hypothesis whose full proof remains open.

Remark 6.5. This is a *compatibilist* position: the laws are deterministic, but genuine choices exist. It differs from both hard determinism (no free will) and libertarian free will (uncaused choices). The choice is *underdetermined* by the cost landscape at decision points—not because the landscape is random, but because it is flat.

Conway and Kochen's Free Will Theorem [7] establishes that if experimenters have free will (their choices are not predetermined by prior history), then so do elementary particles. The RS Gap-45 barrier provides a structural mechanism for this: at decision points, the choice is underdetermined by the ledger history, consistent with the theorem's requirements.

7 The Born Rule from Projection

In standard quantum mechanics, the Born rule ($P = |\psi|^2$) is a postulate. In RS, its structural basis emerges from the fiber structure of the projection map.

The structural basis for the Born rule rests on Corollary 4.7. For any observer with resolution m: the m fibers partition the state space (the sample space of probability theory); the natural measure on each fiber provides a probability for each outcome. A higher-resolution observer (larger m) has finer fibers and can distinguish more states, corresponding to more precise measurements. This provides the sample-space structure of probability without additional postulates.

Remark 7.1 (Status of the Born Rule derivation). Corollary 4.7 shows that any finite-resolution projection partitions the ledger state space into fibers, providing the *sample space* of probability theory without additional postulates. This is the structural skeleton of the Born rule. However, the derivation of the *specific* Born weight $P = |\psi|^2$ requires more: one must show that the RS measure on the fibers (induced by the 8-tick phase weights $e^{ik\pi/4}$, $k = 0, \ldots, 7$) reproduces exactly $|\text{amplitude}|^2$. The full explicit derivation of the Born probability formula from the RS fiber structure and 8-tick epoch is *not* completed in this paper; it is an identified open problem.

The comparison table's claim "Born rule is primitive: FALSE" is therefore properly understood as "the Born rule has a structural basis in the RS fiber structure (Corollary 4.7); the complete zero-postulate derivation of $P = |\psi|^2$ is an open problem."

8 Laplace's Demon Dissolved

Laplace's demon—an intellect that knows the complete state and can predict all futures—is coherent in RS. Reality IS deterministic. But the demon requires access to the full ledger state, not just a projection. For any finite-resolution observer, this access is impossible.

Corollary 8.1 (Laplace's Demon Dissolved)**.** *Combining Corollary 3.3 (unique minimizer) and Theorem 4.3 (lossy projection):*

(i) The ledger state uniquely determines all future states (determinism).

(ii) No finite-resolution observer can access the full ledger state (projection is lossy).

(iii) Therefore: determinism is ontologically true but epistemically inaccessible to any finite observer within the ledger.

Remark 8.2. This dissolves the Laplace demon: determinism does not imply predictability. The demon is logically possible but physically unrealizable for any observer within the ledger.

9 Discussion

9.1 Summary of the Resolution

Claim	Status in RS
"Reality is deterministic"	TRUE: unique J-minimizer at each tick
"Observations are random"	TRUE: projection through finite resolution is lossy
"Quantum randomness is fundamental"	FALSE: it is epistemic (about observation), not ontological
"Bell's theorem rules out determinism"	FALSE: it rules out *local* determinism; RS is global
"Free will is an illusion"	FALSE: genuine decision points exist at Gap-45
"Born rule is primitive"	PARTIALLY RESOLVED: the sample-space structure (fiber partition) emerges without postulates; the full derivation of $P = \|\psi\|^2$ from the fiber measure remains an open problem

9.2 Comparison with Other Interpretations

Bohmian mechanics. Like RS, Bohmian mechanics restores determinism with nonlocal hidden variables. But Bohm introduces a *guiding equation* as an additional postulate. RS derives everything from the single cost functional—no guiding equation is needed.

Many-worlds. Everett's interpretation preserves determinism by having all outcomes occur in different branches. RS avoids ontological proliferation: there is one ledger with one deterministic evolution. The "branches" are the fibers of the projection map—mathematical subsets, not physical worlds.

GRW and collapse models. Collapse theories introduce irreducible randomness via stochastic modifications of the Schrödinger equation. RS requires no such modification: the apparent randomness arises entirely from finite-resolution projection.

't Hooft's cellular automaton. 't Hooft [10] also proposes a deterministic substrate beneath quantum mechanics, arguing that a cellular automaton at the Planck scale can reproduce quantum field theory in an appropriate limit. RS shares 't Hooft's deterministic commitment but differs in two respects: (i) RS derives the discrete structure from the cost functional (the automaton is not postulated but *forced*), and (ii) apparent randomness in RS arises from finite-resolution projection rather than from information loss at the Planck scale.

9.3 Falsifiable Predictions

1. **No irreducible randomness.** If an experiment ever demonstrates randomness that cannot be explained by projection through finite resolution—for example, fun-

damental randomness in a system with access to the full ledger state—the framework is refuted.

2. **Gap-45 decision structure.** The framework predicts that genuine decision points are associated with the Gap-45 periodicity. Neurophysiological experiments could test whether decision timing correlates with period-360 structure.

3. **Born-rule deviations.** At sufficiently high resolution (approaching the fundamental scale), the Born rule should show deviations reflecting the discrete fiber structure.

10 Conclusion

The determinism–randomness debate is dissolved, not by choosing a side, but by proving that both sides are correct about different levels of description:

- **Ontologically**, the universe is deterministic. The strict convexity of the cost functional J guarantees a unique minimizer at every ledger update.
- **Epistemically**, observations appear random. Finite-resolution projection is provably lossy, and the resulting fiber structure reproduces quantum statistics.
- **Agentially**, free will is real. At genuine decision points (flat J-landscape, Gap-45 barrier), the agent must choose among near-degenerate futures, and no external system can predict the choice.

All three levels coexist without contradiction within a single framework derived from one cost functional and zero adjustable parameters.

References

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